

Karl-Alexandre Michaud

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EDUCATION

University of Toronto

Toronto, Ontario, CA

Honours Bachelor of Science, Double Major in Computer Science, Mathematics

Sep 2023 – Jun 2027

- **Courses:** Software Design, Systems Programming, Data Structures, Algorithms, Computer Architecture (Intro)

EXPERIENCE

Software Developer

Sep 2025 – Present

saycamel.com

Toronto, Ontario, CA

- Built a **social platform** with **750+** registered users and **3k+** monthly visitors, connecting students to **1400+** UofT clubs and **240+** weekly campus events using **React**, **JavaScript**, and **Supabase**
- Led a team of 3 to develop the club dashboard and **collaborated** with cross-functional teams to design and deploy **end-to-end features**, supporting **100+** prospective clubs and **increasing engagement by 28%**

Software Developer (Markus, PythonTA)

May 2025 – Present

University of Toronto, Department of Computer Science

Toronto, Ontario, CA

- Engineered and maintained **MarkUs**, UofT's assignment and grading platform for **Computer Science and Statistics**, supporting **10k+** daily student users, leveraging **Ruby on Rails**, **JavaScript**, and **React**
- Contributed to **PythonTA**, a static code analysis tool used in UofT courses to automatically check code style and logic for **35,000+** students since 2016, leveraging **Python** and **Astroid**
- Optimized performance for **PythonTA** by **reducing import time by 9.98%** through **analytical problem-solving** and **refactoring**, improving efficiency for **3900+** annual users

PROJECTS

CExpress | C, Shell, Make, Homebrew

karl-michaud.github.io/CExpress/

- Created **CExpress**, a lightweight open-source **HTTP server / microservice framework** for **C** that enables rapid prototyping and development of **RESTful APIs** with minimal overhead
- Implemented key features such as **dynamic routing**, **HTTP/1.1 request handling**, and low-level **TCP socket programming**, showcasing expertise in **network programming** with an **optimized memory footprint**

NorthBlue Network | Next.js, TypeScript, Supabase

www.northblue.network

- Built a **networking platform** connecting UofT builders, creators, and founders, generating **10K+** impressions in 24 hours and attracting users including **4 VC investors** and notable startup founders
- Implemented authentication, profile discovery, and real-time updates using **Next.js**, **TypeScript**, and **Supabase**

AI Video Generator | C, Python, Shell, FFmpeg, Deepseek API, Whisper

www.youtube.com/watch

- Engineered a fully **autonomous video generation pipeline** in **C**, coordinating script generation, TTS synthesis, subtitle burning, and media merging using **FFmpeg**, **Whisper**, and **DeepSeek API**
- Architected a **multi-process pipeline** of 5 standalone CLI programs communicating through **Unix pipes**

Recyclobot – DeltaHacks 11 | Swift, SwiftUI, Python, Firebase, OpenAI API devpost.com/software/recyclobot

- Built an AI-driven **waste management system** with **real-time object detection** and sustainability insights
- Delivered a functional **iOS app + hardware prototype** in **24 hours**, ranking in the **top 12 overall** and **top 3 environmental** categories among **600+** participants through rapid, iterative development

LEADERSHIP AND INVOLVEMENT

Event Coordinator

Aug 2024 – May 2025

University of Toronto, STEP UofT Student Club

Toronto, Ontario, CA

- Led **event planning** during the club's founding year, organizing **5 events** with **15+** attendees each
- Onboarded **25+** new members, driving club growth through strategic **communication and collaboration**
- Secured **5 UTSL Club Award nominations** in our inaugural year by improving event logistics

TECHNICAL SKILLS

Languages: Java, C, Python, JavaScript, Ruby, Swift, HTML, CSS, SQL, Shell, MIPS Assembly, Makefile

Frameworks / Libraries: Rails, Node.js, Next.js, SwiftUI, Maven, React, Pytest, JUnit, Astroid, Rspec

Developer Tools: Git, GitHub, CI/CD, MySQL, Firebase, Linux, OpenAI API / DeepSeek API, Whisper